A Square-Root-Free Parametrization of 6-Point Kinematics

(four massless legs, two equal-mass legs)

TriangleHeptagon3 – `4DBasis` notes

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1 Setup and conventions

We work with six outgoing momenta $p_1, \ldots, p_6$ satisfying momentum conservation

$$p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 0,$$

with legs 1–4 massless and legs $5, 6$ of equal mass,

$$p_1^2 = p_2^2 = p_3^2 = p_4^2 = 0, \qquad p_5^2 = p_6^2 \equiv m_T^2 \ (\texttt{mt2}).$$

The standard (redundant) set of Mandelstam invariants used throughout the project is

$$\texttt{diffInvariants} = \{\, m_T^2,\ s_{12},\ s_{123},\ s_{16},\ s_{23},\ s_{234},\ s_{34},\ s_{345},\ s_{45},\ s_{56} \,\},$$

with $s_{ij} = (p_i + p_j)^2$ and $s_{ijk} = (p_i + p_j + p_k)^2$. These ten numbers are *not* independent: they satisfy exactly one polynomial relation, stored in `SetUp/SixPointGram.txt` and referred to below as $G_6 = 0$.

2 The problem: the Gram constraint forces a square root

Any five four-vectors in four-dimensional Minkowski space are linearly dependent, so their 5×5 Gram matrix is singular:

$$G_6 \ \equiv\ \det\left(p_i \cdot p_j\right)_{i,j\in\{1,2,3,4,5\}} \ =\ 0.$$

Written out in terms of the ten named invariants, G_6 is a specific polynomial that is *quadratic* in s_{16} (and, in fact, in several of the other invariants too). Solving $G_6 = 0$ for s_{16} therefore gives

$$s_{16} \ =\ R(\text{other 9}) \ \pm\ \sqrt{\Delta(\text{other 9})}$$

for some rational function R and polynomial Δ: an unavoidable square root, since the ten invariants are only a redundant, 9-dimensional description of the true phase space to begin with.

This square root is exactly what the construction below removes: instead of using the ten invariants as coordinates and solving a quadratic constraint among them, we parametrize the momenta directly, so that the constraint is never imposed after the fact – it is automatically satisfied by the parametrization itself.

3 Resolution: writing p_5 as a combination of $p_1, \ldots, p_4$

Generically, p_1, p_2, p_3, p_4 already span the full four-dimensional momentum space. Any fifth momentum is then necessarily a linear combination of them:

$$\boxed{p_5 = c_1\, p_1 + c_2\, p_2 + c_3\, p_3 + c_4\, p_4} \qquad (\star)$$

and momentum conservation fixes p_6 linearly, with no root:

$$p_6 = -(p_1 + p_2 + p_3 + p_4 + p_5).$$

Because $(\star)$ makes $p_1, \ldots, p_5$ linearly dependent *by construction*, for *any* choice of c_1, c_2, c_3, c_4, the Gram determinant $G_6 = \det(p_i \cdot p_j)_{i,j=1}^{5}$ vanishes identically. $G_6 = 0$ is not an extra condition to impose on $(\star)$ – it is a consequence of it. (The actual code computes G_6 as the Gram determinant of $\{p_6, p_2, p_3, p_4, p_5\}$ with p_6 eliminated via momentum conservation; a standard row-operation argument – adding $p_2 + p_3 + p_4 + p_5$ to the first row/column – shows this equals $\det(p_i \cdot p_j)_{i,j=1}^{5}$ up to an overall sign, so the two statements agree.)

4 The shape of $p_1, \ldots, p_4$: a sixth invariant, s_{14}

Four massless momenta are specified by six independent numbers: the pairwise dot products, equivalently

$$s_{12},\ s_{13},\ s_{14},\ s_{23},\ s_{24},\ s_{34}.$$

The named invariants supply five of these:

$$s_{12},\ s_{23},\ s_{34}\ \text{(directly)}, \qquad s_{13} = s_{123} - s_{12} - s_{23}, \qquad s_{24} = s_{234} - s_{23} - s_{34}.$$

There is no representative of $s_{14} = (p_1 + p_4)^2 = 2\, p_1 \cdot p_4$ among $\{s_{12}, s_{23}, s_{34}, s_{123}, s_{234}\}$: it is exactly the piece of information that used to be hidden inside the s_{16} square root. It must be introduced as a genuinely new, independent symbol.

5 Master formulas

Write the dot products of the $p_1, \ldots, p_4$ basis as

$$d_{12} = \tfrac{1}{2}s_{12}, \quad d_{23} = \tfrac{1}{2}s_{23}, \quad d_{34} = \tfrac{1}{2}s_{34}, \quad d_{13} = \tfrac{1}{2}(s_{123} - s_{12} - s_{23}), \quad d_{24} = \tfrac{1}{2}(s_{234} - s_{23} - s_{34}), \quad d_{14} = \tfrac{1}{2}s_{14}.$$

From $(\star)$, every invariant involving p_5 follows by direct expansion. Since $p_i^2 = 0$ for $i = 1, \ldots, 4$,

$$m_T^2 = p_5^2 = \sum_{i,j=1}^{4} c_i c_j\, d_{ij} = c_1c_2s_{12} - c_1c_3(s_{12} - s_{123} + s_{23}) + c_2c_3s_{23} + c_1c_4s_{14} - c_2c_4(s_{23} - s_{234} + s_{34}) + c_3c_4s_{34}.$$

Note the structure: because $d_{ii} = 0$ for all i (masslessness of $p_1, \ldots, p_4$), m_T^2 has no c_i^2 term – it is linear in each c_i individually, even though it is an overall quadratic (bilinear) form in the four of them jointly.

The remaining three dependent invariants are obtained the same way, each as (known massless sum)^2+ 2(known)$\cdot p_5 + p_5^2$:

$$s_{45} = (p_4 + p_5)^2 = m_T^2 + 2\, p_4 \cdot p_5, \qquad (1)$$

$$s_{345} = (p_3 + p_4 + p_5)^2 = s_{34} + 2\,(p_3 + p_4) \cdot p_5 + m_T^2, \qquad (2)$$

$$s_{16} = (p_1 + p_6)^2 = (p_2 + p_3 + p_4 + p_5)^2 = s_{234} + 2\,(p_2 + p_3 + p_4) \cdot p_5 + m_T^2, \qquad (3)$$

using $p_1 + p_6 = -(p_2 + p_3 + p_4 + p_5)$. Finally,

$$s_{56} = (p_5 + p_6)^2 = (p_1 + p_2 + p_3 + p_4)^2 = s_{123} + s_{14} - s_{23} + s_{234},$$

which is *linear*, with no c_i-dependence at all – it is fixed as soon as the shape of $p_1, \ldots, p_4$ is fixed. All four of $m_T^2, s_{45}, s_{345}, s_{16}$ are manifestly rational (in fact polynomial) in

$$\{\, s_{12}, s_{23}, s_{34}, s_{123}, s_{234}, s_{14}, c_1, c_2, c_3, c_4 \,\}.$$

No square root has appeared anywhere.

6 The one real constraint: $p_5^2 = p_6^2$

Nothing in ($\star$) forces leg 6 to have the same mass as leg 5 – that is genuine extra physical input, not a kinematic identity. Expanding,

$$p_6^2 - p_5^2 = (p_1 + p_2 + p_3 + p_4)^2 + 2\, p_5 \cdot (p_1 + p_2 + p_3 + p_4),$$

which is manifestly *linear* in c_1, c_2, c_3, c_4 jointly: the quadratic $c_i c_j$ pieces of p_5^2 and p_6^2 cancel identically between the two. Explicitly,

$$\begin{aligned} p_6^2 - p_5^2 \;=\; & s_{123} + s_{14} - s_{23} + s_{234} + c_1\big(s_{123} + s_{14} - s_{23}\big) + c_2\big(s_{12} + s_{234} - s_{34}\big) \\ & + c_3\big(s_{34} + s_{123} - s_{12}\big) + c_4\big(s_{14} - s_{23} + s_{234}\big) \;=\; 0. \end{aligned}$$

Solving this *linear* equation for c_4,

$$c_4 \;=\; \frac{-s_{123} - s_{14} + s_{23} - s_{234} - c_1(s_{123} + s_{14} - s_{23}) - c_2(s_{12} + s_{234} - s_{34}) - c_3(s_{34} + s_{123} - s_{12})}{s_{14} - s_{23} + s_{234}},$$

an ordinary division – never a square root.

7 The resulting 9-parameter description

Substituting this c_4 into the master formulas of Section 5 leaves $m_T^2, s_{45}, s_{345}, s_{16}$ as rational functions of exactly

$$\boxed{\{\, s_{12},\ s_{23},\ s_{34},\ s_{123},\ s_{234},\ s_{14},\ c_1,\ c_2,\ c_3 \,\}}$$

– nine totally free, unconstrained parameters, with $s_{56} = s_{123} + s_{14} - s_{23} + s_{234}$ and no relation of any kind left to satisfy. This matches the expected counting: the ten named invariants carry one relation ($G_6 = 0$), hence nine independent degrees of freedom; this is simply a different, root-free coordinate system on the same nine-dimensional variety.

In `4DBasis/RationalParametrization.wl` this is implemented as `mt2Formula`, `s45Formula`, `s345Formula`, `s16Formula`, `s56Formula`, `massEqualityConstraint`, `c4Formula9`, and packaged as the substitution list `ReduceTo9Variables` (which also eliminates c_4), together with the sampler `BuildRationalPoint[]`.

8 Verification

Two independent checks confirm the construction:

- **Symbolic:** substituting the reduced formulas into the actual G_6 polynomial from `SetUp/SixPointGram.txt` and calling `Together` gives the identically zero polynomial – not merely zero at sample points, but a true rational-function identity.
- **Numeric:** sampling `BuildRationalPoint[]` many times over the finite field $\mathbb{F}_p$ with $p = 2^{31} - 1$ always lands exactly on $G_6 = 0$ (50/50 and later 10/10 independent trials checked). By contrast, the old method – drawing the other nine invariants at random and solving the quadratic $G_6 = 0$ for s_{16} over $\mathbb{F}_p$ – has *no* solution whenever the discriminant is a quadratic non-residue, i.e. roughly half the time: a direct, concrete manifestation of the square root this construction avoids.

9 The inverse map: recovering (c_1, c_2, c_3, c_4) from a known point

Given a point already expressed in the old, redundant ten invariants (satisfying $G_6 = 0$, e.g. produced the old way), the four dot products $p_5 \cdot p_1, \ldots, p_5 \cdot p_4$ are already fully determined:

$$e_4 = \tfrac{1}{2}(s_{45} - m_T^2), \quad e_3 = \tfrac{1}{2}(s_{345} - s_{34} - s_{45}), \quad e_2 = \tfrac{1}{2}(s_{16} - s_{234} - m_T^2) - e_3 - e_4, \quad e_1 = -\tfrac{1}{2}s_{56} - e_2 - e_3 - e_4.$$

Inverting the (generically invertible) Gram matrix $\left(d_{ij}\right)_{i,j=1}^{4}$ of the basis $p_1, \ldots, p_4$ against (e_1, e_2, e_3, e_4) gives a *unique* (c_1, c_2, c_3, c_4) – there is no leftover freedom, since a point on $G_6 = 0$ pins p_5 down completely. This is implemented as `CoefficientRulesFromInvariants` / `CoefficientsAt[ ]` and verified to round-trip exactly (`VerifyInverseMap[ ]`).

This construction was independently discovered, for the *same* external kinematics (identical `mandelstam_scalarproducts.txt` conventions), in a different loop-topology project's `sector239Basis.wl` (`kinematicRules239`). The c_1, c_2, c_3 obtained there were checked to agree with the ones derived here *as exact rational-function identities*; that file omits c_4 because it never needs it on its own, not because it is somehow absent.

A simplification. Since $p_4^2 = 0$, $m_T^2 = p_5^2$ is already *linear* in c_4 alone, so c_4 can be read off directly from $m_T^2 = p_5^2$ (holding c_1, c_2, c_3 and the named invariants fixed) instead of inverting the full 4×4 system:

$$c_4 \;=\; \frac{-m_T^2 + c_1c_2s_{12} - c_1c_3s_{12} + c_1c_3s_{123} - c_1c_3s_{23} + c_2c_3s_{23}}{c_1s_{123} - c_1s_{23} + c_2s_{23} + c_1s_{234} - c_2s_{234} + c_2s_{34} - c_3s_{34} - c_1s_{56}},$$

dramatically shorter than the corresponding entry obtained by Cramer's rule on the full system, and checked numerically to agree with it on many independent finite-field points.

10 Alternative bases: trading a shape invariant for m_T^2

In the 9-parameter set of Section 7, m_T^2 is a *derived* quantity – a specific function of the nine free parameters. It is natural to ask whether m_T^2 can instead be one of the free parameters, with something else derived. Checking the degree of (the reduced) m_T^2 in each of the nine:

c_1, c_2, c_3, s_{14}, s_{23}, s_{34}, s_{234}	degree 2
s_{12}, s_{123}	degree 1

Only s_{12} and s_{123} sit at degree 1: these are the *only* two that can be traded for m_T^2 without reintroducing a square root. In particular c_4 was *never* a candidate for this swap – it is not one of the 9 free parameters to begin with, having already been eliminated by the unrelated mass-equality constraint of Section 6 – and s_{14} cannot be traded either, despite being the "extra" invariant introduced in Section 4: it is genuinely quadratic in the m_T^2 relation, exactly like $c_1, c_2, c_3, s_{23}, s_{34}, s_{234}$.

This gives two further valid, fully rational 9-parameter bases:

$$\{\, m_T^2,\ s_{23},\ s_{34},\ s_{123},\ s_{234},\ s_{14},\ c_1,\ c_2,\ c_3 \,\} \qquad (s_{12} \text{ dropped}),$$
$$\{\, m_T^2,\ s_{12},\ s_{23},\ s_{34},\ s_{234},\ s_{14},\ c_1,\ c_2,\ c_3 \,\} \qquad (s_{123} \text{ dropped}),$$

implemented as `ReduceTo9VariablesKeepMt2` and `ReduceTo9VariablesKeepMt2AndS12` respectively, each verified the same way as Section 8 (exact symbolic vanishing of G_6), together with matching flat kinematic-rule files `kinematicRulesTH3KeepMt2.txt` and `kinematicRulesTH3KeepMt2AndS12.txt`.

11 Summary

Free 9-parameter basis	Derived quantities	File object
$s_{12}, s_{23}, s_{34}, s_{123}, s_{234}, s_{14}, c_1, c_2, c_3$	$m_T^2, s_{16}, s_{45}, s_{345}, s_{56}, c_4$	`ReduceTo9Variables`
$m_T^2, s_{23}, s_{34}, s_{123}, s_{234}, s_{14}, c_1, c_2, c_3$	$s_{12}, s_{16}, s_{45}, s_{345}, s_{56}, c_4$	`ReduceTo9VariablesKeepMt2`
$m_T^2, s_{12}, s_{23}, s_{34}, s_{234}, s_{14}, c_1, c_2, c_3$	$s_{123}, s_{16}, s_{45}, s_{345}, s_{56}, c_4$	`ReduceTo9VariablesKeepMt2AndS12`

In every case: nine free, unconstrained parameters; zero square roots; $G_6 = 0$ holds as an algebraic identity, not an equation left to solve.